Chapter 3, Problem 2RQ

Problem

In the circuit of Fig. 3.46, applying KCL at node 2 gives:

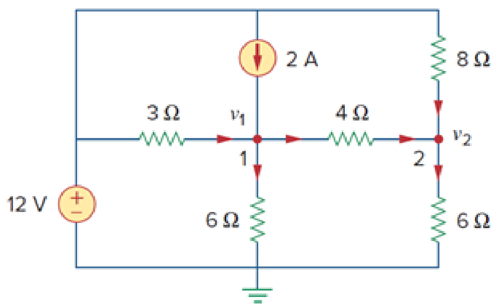
(a) $\dfrac{v_2 - v_1}{4} + \dfrac{v_2}{8} = \dfrac{v_2}{6}$

(b) $\dfrac{v_1 - v_2}{4} + \dfrac{v_2}{8} = \dfrac{v_2}{6}$

(c) $\dfrac{v_1 - v_2}{4} + \dfrac{12 - v_2}{8} = \dfrac{v_2}{6}$

(d) $\dfrac{v_2 - v_1}{4} + \dfrac{v_2 - 12}{8} = \dfrac{v_2}{6}$

Figure 3.46:



Step-by-step solution

Step 1 of 1

Answer: (c)

$$\frac{V_1 - V_2}{4} + \frac{12 - V_2}{8} = \frac{V_2}{6}$$